Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

ME 255 – Strength of Materials (Mechanics of Materials)

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LECTURE 1

Introduction
&
Recap of Static Mechanics of Solids

Design

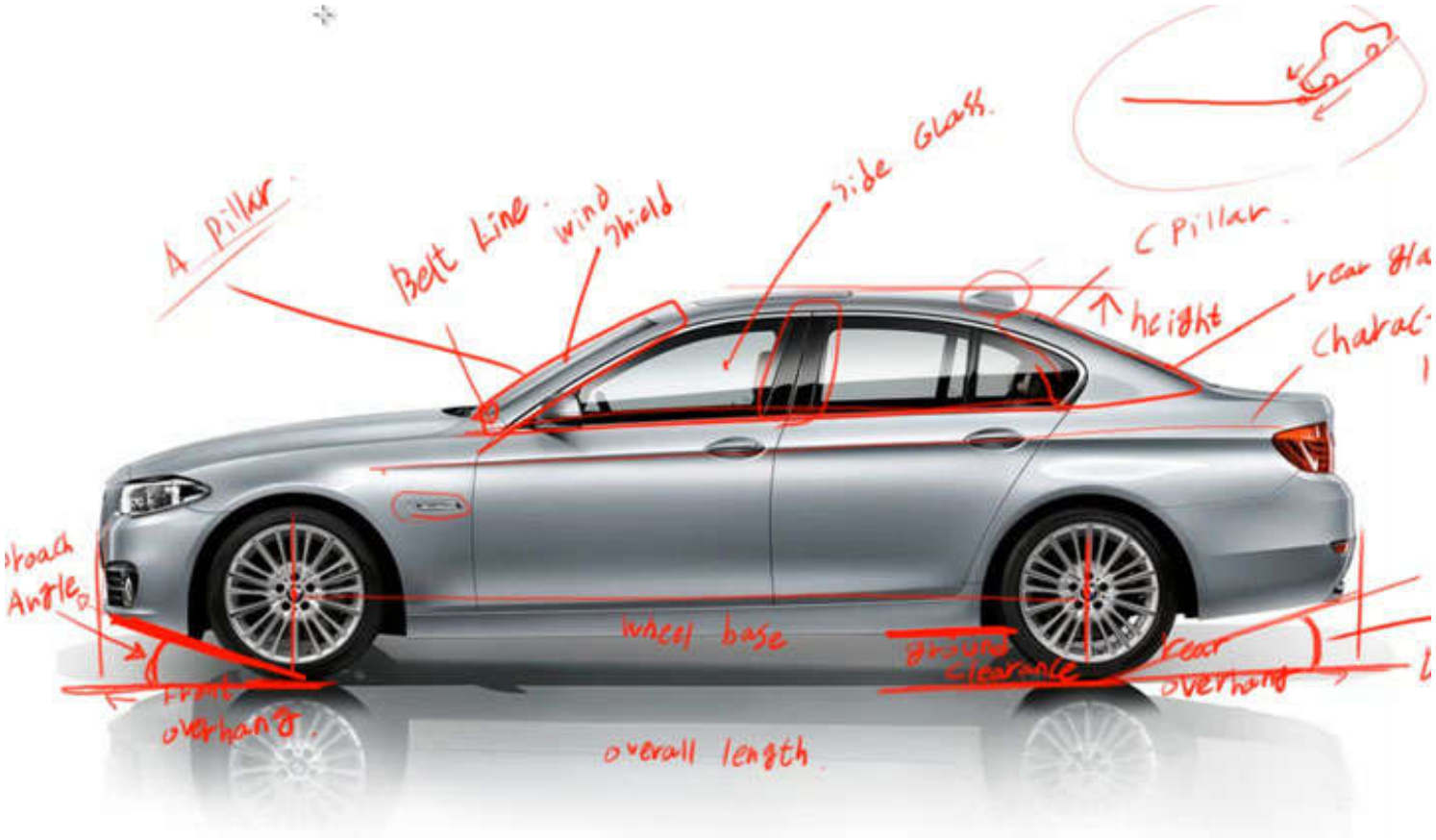
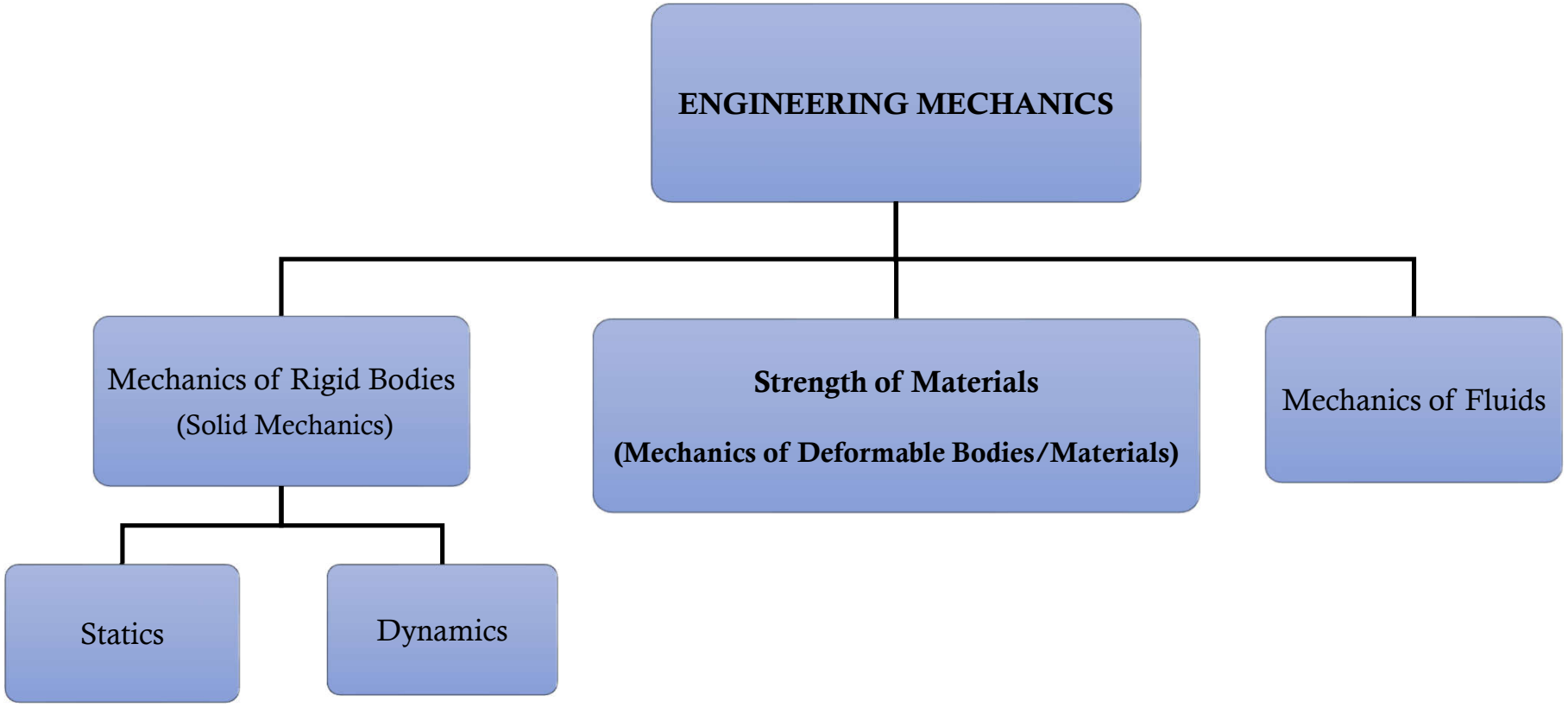


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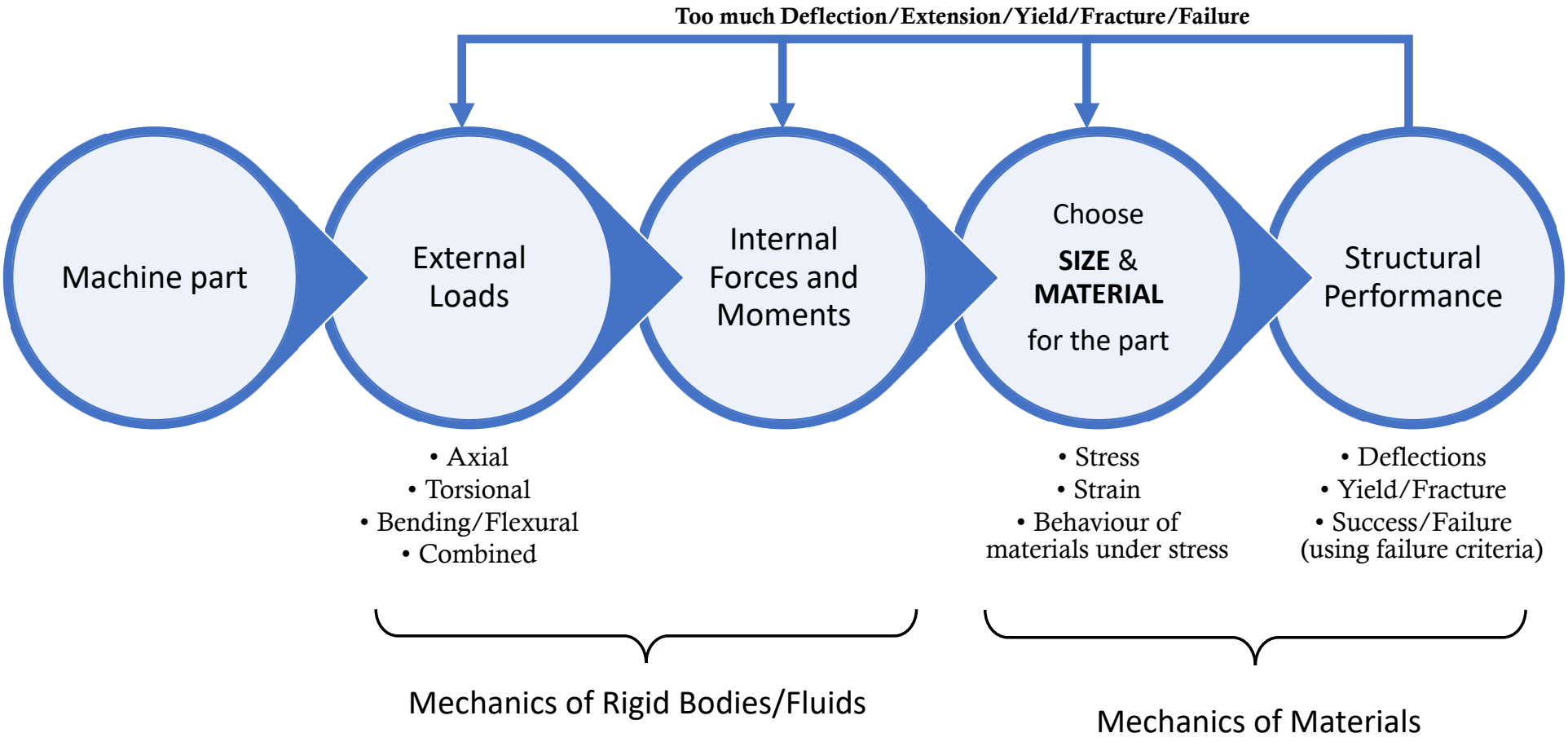
- Selecting from available choices to meet desired expectations → ***designing***.

Branches of Engineering Mechanics

Some Branches of Engineering Mechanics
(J. Antonio)



Generic process for designing an engineering structure



Outline

- ❑Review of Principles of Static Equilibrium
- ❑Stress and Strain under Different Loads
  - ❑ Normal (Direct) Stress and Strain (from Axial loads)
    - Normal Stress and Normal strain
    - Shear stress and Shear strain
    - Stress concentrations
    - Mechanical Properties of Materials; Stress - strain Relationships, Hooke's Law, Poisson's Ratio
    - Elastic Deformation under axial loading and its application in Statistically Indeterminate Structures.
    - Mohr's Stress and Strain Circles.
  - ❑ Thermal stress and Strain.
  - ❑ Torsional loads;
    - Torsional Stress and Strain;
    - Polar Moments of Inertia;
    - Elastic deformation under torsional loads
    - Stress Concentrations
  - ❑ Bending loads;
    - Shear force and bending moment diagrams,
    - Centroids and Area Moments of Inertia
    - Flexural Stress in elastic region
- ❑Combined loading and Theories of Static failure
- ❑Applications in Thin-Walled Pressure Vessels

Leaning Outcomes

- Calculate simple stresses and strains of structures within the elastic limit.
- Understand and calculate torsional stresses and strains in circular solid and hollow shafts
- Draw shearing force and bending moment diagrams for beams.
- Understand and calculate bending/flexural stresses simple beams.
- Understand and calculate thermal stresses and strains in simple structures.
- Understand different types of loading and apply Mohr's stress and strain circles.
- Understand and apply the theories of static failure to compute failure stresses and strains, and size member under simple loading.

Review Of Some Concepts From Statics Of Rigid Bodies

Mechanics of Static Rigid Bodies

- Concepts of particles, bodies, space, time, mass.
- Basic and derived quantities.
- Forces; characteristics of forces, adding forces (resultant forces), moments.
- Static Equilibrium of particles and bodies.
- Applications of Static Equilibrium.

Conditions of Static Equilibrium

- A particle or body is said to be in equilibrium if the resultant force and moment acting on it is zero. In other words, the sum of forces (or moments) **must** be equal to zero.

- For particles;

$$\vec{R} = \sum F = 0$$
$$\Rightarrow \sum F_x = 0 \qquad \sum F_y = 0 \qquad \sum F_z = 0$$

- For bodies;

$$\vec{R} = \sum F = 0$$
$$\Rightarrow \sum F_x = 0 \qquad \sum F_y = 0 \qquad \sum F_z = 0$$
$$\vec{M} = \sum M = 0$$
$$\Rightarrow \sum M_x = 0 \qquad \sum M_y = 0 \qquad \sum M_z = 0$$

Static Equilibrium Analysis

- Involves three main steps;
  - ✓Sketch a free body diagram for the problem
  - ✓Sum up forces and moments acting on the body to obtain the **equations of equilibrium** for the problem.
  - ✓Solve the equations and interpret your results.

Free Body Diagrams

- Select the extent of the body that is of interest, detach it from the ground and all other bodies and supports, and (basically) sketch the outline of the detached (free) body.

- Indicate force and moment reactions that are exerted on the "free-body" by the ground and other supports that keep it in equilibrium (satisfying Newton's 3rd Law of motion)

- Indicate external forces and moments, including the rigid body weight where it cannot be ignored at their points of application.

- Include the required dimensions to compute the moments of the forces where necessary.

Free Body Diagrams: Support Reactions

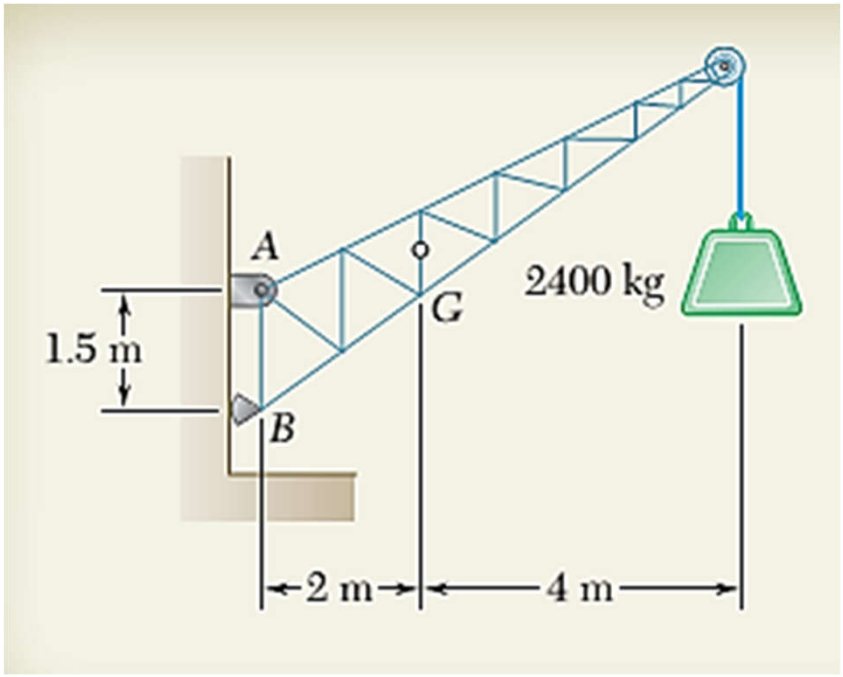
➢Reactions at Supports and Connections for Two-Dimensional Structures

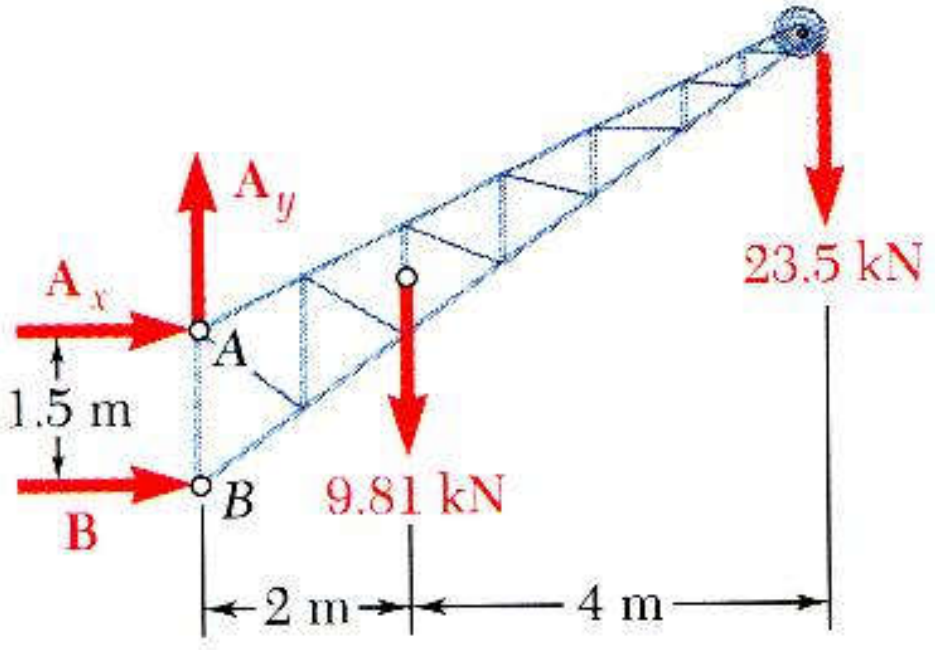
Support or Connection	Reaction	Number of Unknowns
Rollers; Rocker; Frictionless surface	Force with known line of action	1
Short cable; Short link	Force with known line of action	1
Collar on frictionless rod; Frictionless pin in slot	90°; Force with known line of action	1
Frictionless pin or hinge; Rough surface	or; α; Force of unknown direction	2
Fixed support	or; α; Force and couple	3

- Reactions equivalent to a force with known line of action.
- Reactions equivalent to a force of unknown direction and magnitude.
- Reactions equivalent to a force of unknown direction and magnitude and a couple of unknown magnitude

Source: Vector Mechanics for Engineers, Beer *et al.*

Static Equilibrium of Rigid Bodies

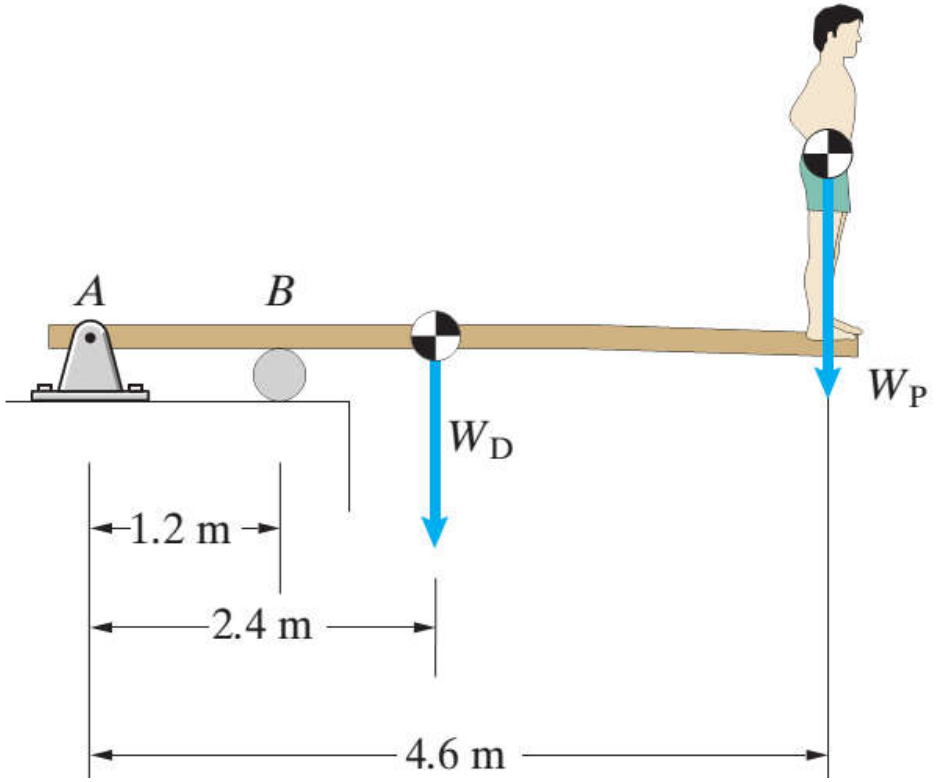




Static Equilibrium of Rigid Bodies

The masses of the man and the diving board are 54 kg and 36 kg, respectively. Assume that they are in equilibrium.

a. Sketch the free-body diagram of the diving board.
b. Determine the reactions at the supports A and B.



Static Equilibrium

Example

A 150 N crate rests on the 100 N pickup tailgate as shown. Calculate the tension T in each of the two restraining cables, one of which is shown. The centres of gravity of the gate and crate are at G_1 and G_2 respectively. Assume the crate is located midway between the two cables.

